

Recent advancements in AI have led to the development of sophisticated yet often opaque tools capable of understanding and

generating text, images, and responses across both textual and visual domains. Researchers usually categorize these advanced deep learning AI models based on their learning approach, purpose, and underlying architecture. Large Language Models (LLMs) are models designed to understand and generate human-like text. These models perform tasks such as text generation, summarization, translation, and question answering. Visual Question Answering (VQA) is a task in AI where a model answers natural language questions based on an image. It combines vision and language understanding to interpret and generate responses. Multiple Large Language Models (MLLMs) are advanced AI models capable of processing and reasoning over multiple types of data (e.g., text, images, audio, video). They extend the capabilities of LLMs by integrating vision, speech, or other sensory inputs. MLLMs [11, 33] exemplify this evolution, pushing the boundaries of how AI can interact with and interpret complex data across modalities. The variety of handled tasks, specialized systems and corresponding AI assistants have led to the introduction of the generic term "foundation model" by the Stanford Institute for Human-Centered Artificial Intelligence's (HAI) Center for Research on Foundation Models (CRFM)[7].

Although these models perform well in various tasks, their internal representation and understanding of fundamental concepts, such as color, remain largely unexplored. However, benchmarks devised to evaluate VQA models systematically propose questions on colored objects appearing within the image. Despite their capabilities, these systems often function as "black boxes," providing limited transparency regarding the mechanisms driving their outputs [17]. While demystifying their intricate workings is a long-term goal, a more immediate challenge lies in understanding how these models conceptualize simple yet foundational concepts, such as color. Gaining insight into this aspect is essential for leveraging their potential to generate precise, meaningful, and valuable outputs [24].

In data visualization, color plays a pivotal role in representing, distinguishing, and communicating complex insights to human viewers, enhancing clarity, readability, and interpretability [38]. Investigating how foundation models interpret and structure color knowledge is crucial for unlocking their potential to generate accurate and meaningful visualizations. Natural language interactions (NLI) between humans and computers are already a reality, including in data exploration and visualization tools like Power BI (Microsoft), Tableau (Salesforce), and Looker (Google). Specifying a color by its name for AI-assisted diagram generation would be intuitive, requiring the capability to produce a digital color representation. Requests, expressed in natural language, for changes to colors appearing in diagram images will also need to demonstrate the ability to identify color names from digital color representations. Hexadecimal color values if the source file format is svg for example, or directly from the image for bitmap files.

To gain knowledge about foundation models' color understanding, we propose **Foundation Model ColorMap**, a novel two-step framework designed to extract and visualize models' internal representations of color. As foundation models are inspired by neural processes in the human brain [12], we operate within a stimulus-response paradigm (question, response analysis), offering a chance to systematically investigate their responses and assess their internal representations [28]. The framework comprises two key

components: (1) a protocol for systematically probing models' internal representations of color through strategic color sampling, tailored prompts, and response analysis; and (2) a novel visualization technique that reveals and maps the structure of their color understanding. Together, these components enable the comparison and evaluation of foundation models based on their conceptual integration of color. The method applies to natural language processing models (LLM) in a text-to-text manner and to computer vision models in an image-to-text protocol. By offering a methodology to extract and visually represent color knowledge, this research establishes a foundation for assessing foundation models' capabilities in the domain of color comprehension, with broad implications for data visualization and AI explainability.

2 Previous Works

In this section, we describe previous works concerned with the presented article, about color naming, foundation models and their evaluation.

2.1 Colors and Named Colors

Researchers have studied the concept of color from various perspectives for a long time. Exploring this concept reveals a range of complex issues rooted in the ambiguous nature of color perception and description. Cultural and linguistic differences shape the way colors are named and understood, with each language providing different levels of precision, overlapping, and categorization.

When examining color names as categorical entities, we notice that they often overlap or nest within each other, adding another layer of complexity. The research by Berlin and Kay [21] introduced the idea that color naming follows universal stages in human languages. Their works, known as the World Color Survey (WCS) involved speakers of 110 languages asked to identify colors. The survey used a standardized set of 330 color samples, and researchers displayed the results on a diagram known as the *Munsell Color Chart*. Researchers have also studied new clustering methods in the various color spaces [16]. Notably, the quest for a minimal common set of color names has led to the concept of universal focal points [30]. An online color survey [22] asks users to name various displayed colors corresponding to hexadecimal values, has resulted in a dataset containing English-language color naming for 954 color values which was the used to derive color naming process [5, 13]. This paper contributes to the broader effort of analyzing color usage and the color knowledge that AI models acquire through different training sets and architectures.

2.2 Foundation Models

Artificial intelligence systems are advancing at an unprecedented pace. The taxonomy of the neural network based systems has evolved in parallel. As defined by Bommasani *et al.* from the Center for Research on Foundation Models (CRFM), a "foundation model is any model that is trained on broad data (generally using self-supervision at scale) that can be adapted (e.g., fine-tuned) to a wide range of downstream tasks". The development of Natural Language Processing (NLP) and transformers [19, 37] leads to the apparition of Large Language Models (LLM) giving way to a range of models designed for diverse applications in AI [18, 25, 35, 36] and to multi

purpose models (MLLM) handling image understanding and creation but also text analysis and generation are called multimodal systems or Multimodal Large Language Models (MLLM) [11, 33] (see [23] for an overview). However, some models directly target image processing and question answering, such as CLIP [27] and OWL-ViT [4] which can be grouped as "visual foundation models".

Research in image understanding continually advances the performance of foundation models and a lot of the recently published foundation models are multimodal.

2.3 Evaluation of Foundation models

Evaluating foundation models involves assessing their performance across various dimensions, such as reliability, accuracy, coherence, fluency, and contextual understanding. This process typically includes both quantitative and qualitative methods [29] and the development of metrics [26]. In 2023, Fu *et al.* [10] published the results of several multimodal models evaluation. Researchers have proposed various benchmark evaluations [8, 9, 23].

Focusing on colors in neural network based AIs and based on the Munsell Chart colors, Abdou *et al.* [1] studied the perceptual structure encoding of LLMs, finding alignment between model representation extraction data of a rather small set of color names, extracted from the Color Lexicon of American English, and Lab color values. Salin *et al.* have published another research [32] which compares pre-trained and fine-tuned representations for vision, language and multimodal aspects. They evaluated, among other aspects, color understanding, using 8 common colors: blue, red, black, white, yellow, orange, green, purple. Boukhers *et al.* [3] have produced a benchmark dataset of 23,469 tuples (Image, Question, Answer). Their work is focusing on realistic images, colors and shapes. A neural network-based semantic system for assigning colors to text has also been proposed [6].

Concerning image understanding by multimodal models, researchers have produced benchmarks as the proposal by Zhang *et al.* [39] which is oriented toward qualitative information like attribute classification, got for example by the following prompt: "What color are her eyes?". In 2020, Zou and Xie published a survey on Visual Question Answering [40] that reviews and analyzes existing datasets, metrics, and models. Researchers [2, 15] created dedicated visual question answering probing datasets.

Highlighting the significance of graphic design topics in AI, researchers proposed a survey of graphic design intelligence in 2023[14]. This suggests that we need more research to better understand how foundation model systems comprehend color. To the best of our knowledge, no previous work investigated, with the depth we present, how to first retrieve color knowledge data and to represent it. This work is an attempt to fill this gap.

3 Extracting Foundation Model Color Knowledge

Previous studies on color knowledge in foundation models emphasize the need for further research to precisely analyze color comprehension across different foundation models. In this paper, we introduce the protocol we designed to tackle this task and explore various aspects of color-related tasks that a digital assistant could handle.

We focus on the aspects of our protocol related to color knowledge extraction in this section, while section 4 details the exploitation of the results.

Section 3.1 discusses the selection of the color space used to obtain color samples for testing. Section 3.2 then focuses on the core subject: color name identification. In paragraph 3.2.1, we describe the extraction process designed for digital text assistants, followed by a detailed explanation of knowledge extraction for vision-specific models in paragraph 3.2.2.

3.1 Choosing Color Spaces

Extracting a model's color knowledge requires selecting appropriate color spaces for result visualization, color sampling, and model prompting. For prompting, the color information must ideally appear consistently in a single textual representation format in the learning datasets. This ensures its integration within the model and facilitates the parsing of the foundation model's response. Typically, RGB triplet text representations employ parameters within the range of [0, 255], but sometimes they use the range [0, 1]. Hexadecimal color values (without transparency) appear to be the most suitable form for prompting. Regarding the communication of the result analysis, it is important to note the well-known disadvantages of 3D diagrams, such as object occlusion, difficulty in perceiving the mapping from data to 3D locations, and challenges in volume perception. Diagrams displayed using RGB, HSL, and CIE LCH color spaces are not suitable because of their lack of perceptual uniformity. Two pairs of colors, (c_1, c_2) and (c_3, c_4), that have the same Euclidean distance between their respective colors in RGB space may still be perceived as significantly different by a human. The Lab color space, being perceptually more uniform and widely used by color specialists, is more suitable. Therefore, following the work of Heer *et al.* [13] and other researchers, we base the result display aspect of our protocol on the Lab color space, filtered using gamut clipping. We also perform color sampling in the Lab color space to explore the color volume perceptively and ensure coherence with the display diagrams.

3.2 Associating a Color Name to a Hexadecimal Color Value

This section explains the process of extracting from the model the color name associated with a given hexadecimal color value, dividing the methodology into two distinct approaches.

- (1) A specialized approach for textual foundation models to evaluate the model's capability to generate accurate color names based on hexadecimal color codes.
- (2) An approach focused on visual foundation models and multimodal models, examining the models' performance in naming colors within images.

The two protocols give the ability to compare the models' knowledge in case of color identification, both in textual and visual contexts, to evaluate their accuracy and flexibility.

3.2.1 Textual foundation model, Hexadecimal Color Value to Color Name. There are several approaches to determining whether a foundation model can recognize a color based on its hexadecimal value. A first method may be based on *open-ended* questions in

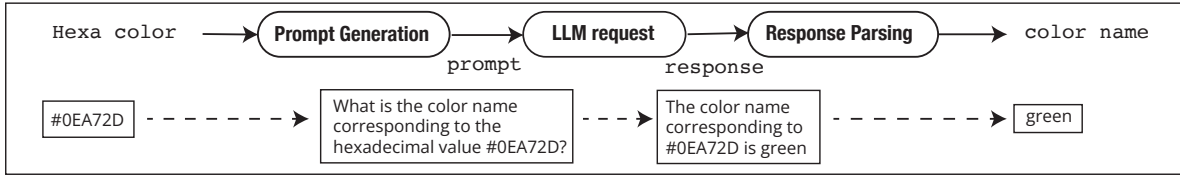


Figure 2: Extracting the color knowledge of a textual foundation model, with the identification of a color name given a hexadecimal color value, the protocol involves several steps: prompt construction, model request, and response parsing.

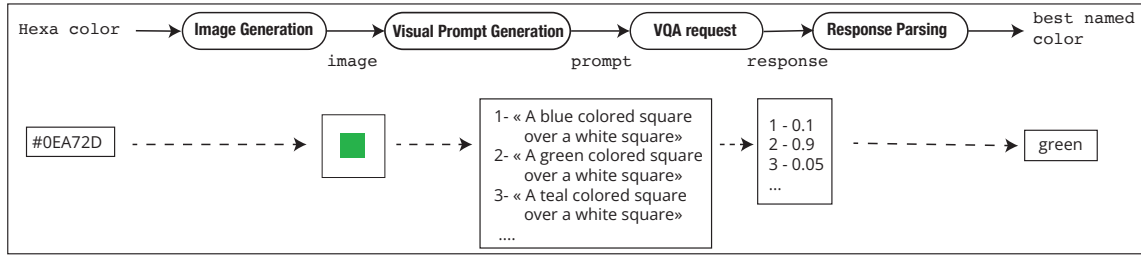


Figure 3: For vision foundation model, to extract the color knowledge concerning the color naming of a hexadecimal color value, we have to generate a picture using the hexadecimal color value, request the model using a dedicated MCQA prompt, parse the response to get the color name with the best rank.

the prompt, such as "What is the name of the color corresponding to the hexadecimal color value #00FF00?". This approach allows the model to interpret and freely provide its best match for the color code, potentially using modifiers like "dark red" or "greenish blue". Another prompt style, called Multiple Choice Question Answering (MCQA), involves presenting the model with a question and several answer choices. It must then select the correct one. This task evaluates the model's ability to understand the question and context, to reason among multiple answers. A corresponding prompt might ask the model to choose from a predefined set of color names, like "Which of the following color names matches the hexadecimal color value <#00FF00>: Red, Green, or Blue?". We can also change the prompt's presentation, using Multiple Choice Prompting (MCP) [31], leading to: "Which of the following color names matches the hexadecimal color value<#00FF00>: 1: Red, 2: Green, 3: Blue? Answer by the corresponding category index." This approach can help in narrowing down the model's response to a specific selection of color names. Figure 2 shows the depiction of this text oriented "Prompt-Request-Parsing" process.

3.2.2 Vision Foundation model: Naming the Color in an Image. As mentioned above, vision foundation models respond to request about images like Vision Question Answering (VQA) tasks. In order to test such models, images containing a colored primitive are generated, and the model is prompted with, various captions mentioning color names for CLIP like models [27] (Contrastive Language Image Pretraining) and with various color descriptors for models like OWL-ViT [4] (Oriented Window Learning Vision Transformer). Finally, the returned ranked answers give the identified color name. These two types of vision-oriented models inherently rely on predefined color names and do not support open-ended prompts.

4 Exploiting extracted color knowledge, models evaluation and comparison

When the extraction part is over, the resulting data comprises a list of the hexadecimal color codes (corresponding to the Lab color space samples) associated with each color name. Analyzing the data to communicate on the foundation model color abilities is mandatory. In the next section, we introduce the diagram types proposed for displaying results and the available display options. Then, in Section 4.2, we discuss some results and demonstrate that the protocol can be extended to Munsell Charts.

4.1 Foundation ColorMap Construction

The hexadecimal color values correspond to points location on the constant luminance Lab plane. The point distribution for a given color name may be dense, sparse, possibly containing outliers. Displaying a surrounding polygon for the obtained locations does not accurately represent the results, as they would be influenced by outliers. To effectively communicate the results, it is more efficient to compute a contour that encloses a zone containing the majority of the 2D locations, and to provide the ability to display the hexadecimal locations on the Lab plane.

To compute a dense point region for a given color name, we use kernel density estimation (KDE)[34] (see figure 4) following the approach proposed by Mayorga *et al.* [20] for scatterplots visualization. Classically, we calculate a surrounding rectangle, expand it and cut it into small squares. For every Lab color point, we add a 2D Gaussian kernel distribution (the kernel's bandwidth is a parameter), creating a density estimation per square.

Then, to get an estimation of the color name zone, we calculate the iso-value 2D curve for a given density value, here half the maximum density (parameter) encountered. Displaying the contours on a constant luminance Lab diagram results in our "Foundation

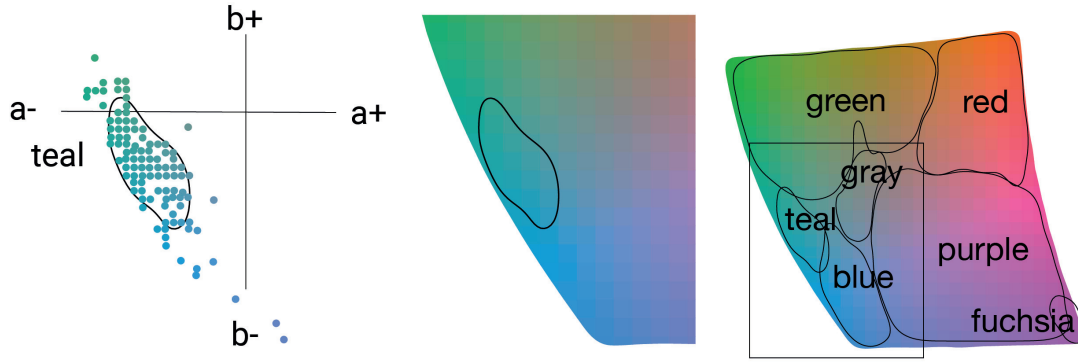


Figure 4: Left and middle: colored discs showing color and location of hexadecimal color values identified as teal color by gemma2 model and corresponding KDE contour. Right: corresponding results for gemma2, MCP prompt on 16 CSS basic color names.

ColorMap“ as visible in figure 1. In the interactive application, the user may display all the data for all the available colors name or only some of them. The contours and corresponding samples may be shown to get a better understanding of the point spreading. The contour may also be filled with the average color of the identified color values.

In addition, we use a “grid diagram” (see figure 6) which displays identified color hexadecimal values as small squares scaled using the number of hexadecimal color values to ensure area equality among color names. The grid diagram minimizes color perception bias that may arise in the Lab display diagram.

4.2 Results and Color Name Recognition

Concerning image understanding in vision foundation models, the results got on CLIP and OWL-ViT show variations in the process of naming color in images. The experiments were using images showing a white background with a centered colored square. Several figures presenting Lab display diagrams, generated for a common Lab luminance to allow comparison, show the ability of our protocol to extract and communicate the color identification model’s knowledge:

- the figure 5 using vision foundation CLIP model and OWL-ViT model, both using prompt based on the color names of CSS basic colors;
- the left and middle parts of figure 1 using LLM models, respectively Llama-3.1-8B-Instruct-Q4_K_M and AIVA/Gemma2-9B-it models, using MCQA prompt providing the color names of CSS basic colors;
- the right part of figure 1 for an open-ended prompt using LLM mistral-7B-Q4-v0.2 model.

The 16 color names of the CSS basic list are: *aqua, black, blue, fuchsia, gray, green, lime, maroon, navy, olive, purple, silver, teal, white and yellow*. A researcher may easily detect questionable associations in the left part of figure 5 showcasing a notable misinterpretation of the color “black”.

Figure 6 shows a “grid” display, corresponding to the Lab display depicted in the middle part of figure 1, using 799 Lab color samples for Lab luminance 60, an MCP prompt based on CSS 16 basic color

names and a AIVA/Gemma2-9B-it LLM model. We discarded the color names lime, navy, olive, maroon, and yellow because they lacked sufficiently recognized hexadecimal values. The grid display diagram helps to compare models without the influence of the surrounding colored zones present in the Lab display.

The protocol can apply to Munsell Chart colors, by prompting the models using the hexadecimal color values of the 330 chips. Two plots, visible in figure 7, allow for a results comparison between CLIP and Gemma2 models. It seems natural that CLIP which is dedicated to image understanding, gives more accurate results than text dedicated models.

Since color naming is influenced by strong cultural factors and lacks a universal consensus or definition for hues and shades corresponding to colors’ names, validation must currently be conducted by humans.

The online XKCD color survey [22] asked participants to name displayed RGB colors, with the results publicly available. The analysis, which derives a single RGB value from all responses for a set of 954 color names, could aid in validating the contours calculated by AI models. We could verify whether each representative RGB value falls within the corresponding color name zone or assess the entire set of user responses.

5 Conclusion

In this paper, we introduced a novel protocol to evaluate foundation models’ deep understanding of color. Composed of two phases, it performs first an extraction of the color knowledge and, in a second phase, analyzes the resulting data before presenting it in various diagrams.

The proposed protocol addresses color name recognition in both images and hexadecimal values. This method involves sampling colors, prompting models, analyzing their outputs, and comparing results through visual diagrams. We uniformly sample the Lab color space at constant luminance to provide hexadecimal values to the model, which then returns corresponding color names. The protocol also demonstrates its applicability to Munsell Chart colors. The result exploiting part, which will be available through a web application, displays color name contours using diagrams of

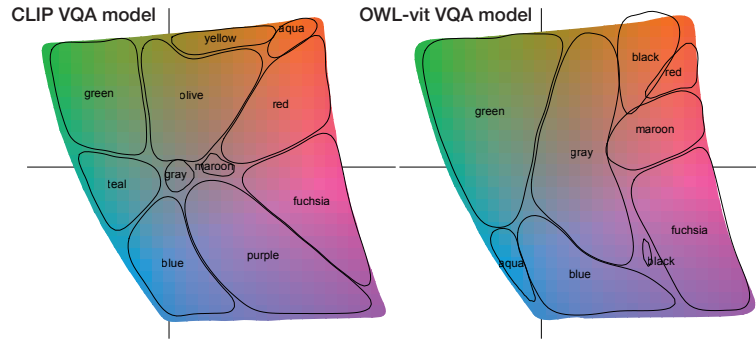


Figure 5: The figure shows on Lab display diagrams the result of two “Visual Question Answering” models color name identification based on hexadecimal values got from constant Lab luminance 60 uniform, 120 samples per axis and images created as “a centered colored square over a white background” (via SVG generation). We restricted the color choices to the CSS 16 basic color names. Left: using OWL-ViT VQA model, and Right: VQA CLIP model. The difference in classification is clearly visible and some color zones are inaccurate, like *black* in CLIP analysis.

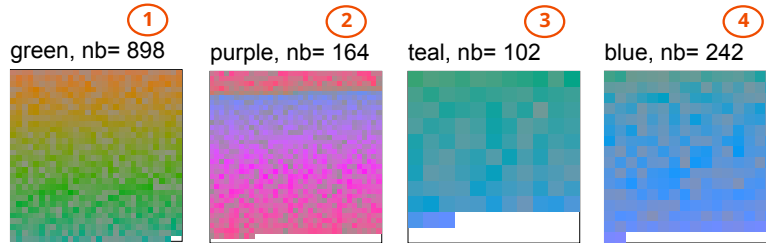


Figure 6: This “Grid” display presents an extract of the hexadecimal color value recognition result, which provides a different perception than the Lab display. The depicted colors result correspond to an extract from AIVA/Gemma2-9B-it model, visible as a Lab diagram in the central part of the figure 1. Some colors hue of square 1 and 2 are inaccurate, like the orange on top of square 1 and the red part in square 2. On the contrary, the presented hues in square 3 and 4 correspond to what we could reasonably expect.

constant luminance Lab color planes, got by KDE densities evaluation. In addition, a grid display, based on squares associated with color names filled with resulting hexadecimal color values, allows for visual inspection of hexadecimal values identified by the color name recognition process.

It extends to vision foundation models by generating images based on hexadecimal color values and querying the models to identify the color name present in the image. Preliminary experiments using different quantizations, models, and prompt types demonstrated the effectiveness of this approach. The presented work is a first step toward a larger analysis of the color concept understanding of foundation models.

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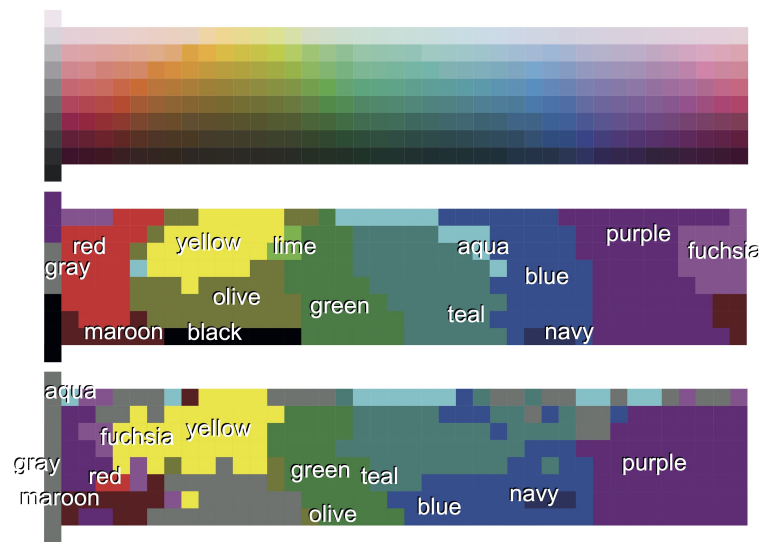


Figure 7: Top: Munsell chart base showing color chips. Middle: Munsell Chart Result using CLIP for Multiple Choice Prompting (MCP) on CSS 16 basic colors names. Below : Munsell Chart Result using Multiple Choice Prompting (MCP) on Gemma2 for CSS 16 basic colors names.

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